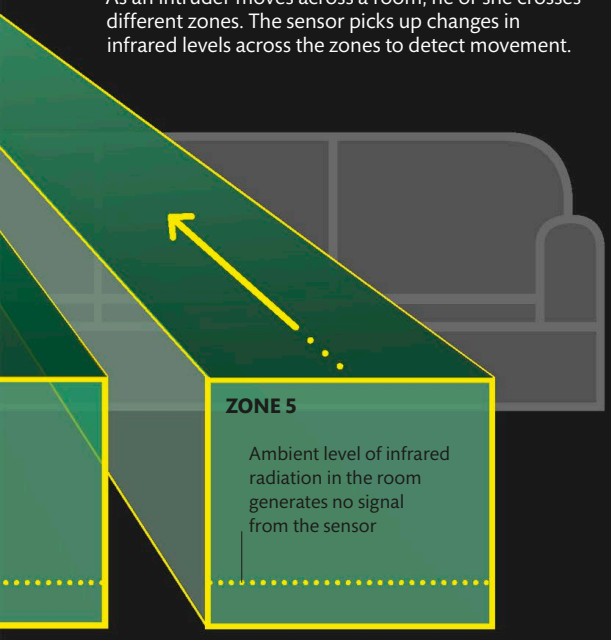
Passive infrared sensors

Everyone gives off infrared radiation at differing levels to their surroundings. Passive infrared (PIR) sensors detect changes in infrared emissions using thin layers of pyroelectric film. This film absorbs infrared radiation, which causes it to heat up and generate small electrical signals. A change in infrared levels across multiple areas of a room can signal the presence and movement of an intruder.

Movement detection
As an intruder moves across a room, he or she crosses different zones. The sensor picks up changes in infrared levels across the zones to detect movement.

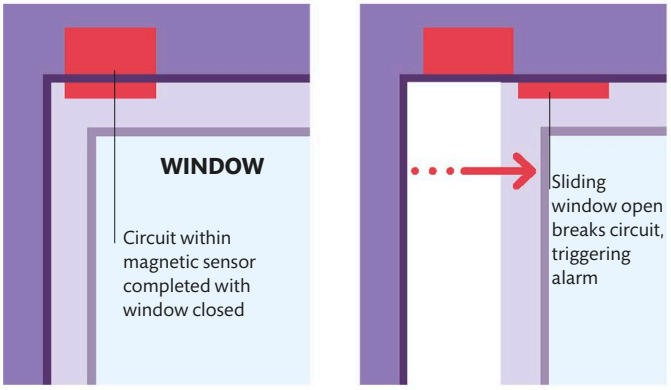


WHERE IS THE BEST PLACE TO PUT A SECURITY SENSOR?

Choke points, such as hallways, where people have to pass through, are good locations, as are corners of rooms offering coverage of multiple entry points.

Contact sensors

The two parts of a magnetic contact sensor—one part fitted to a door or window, the other to its fixed frame—complete an electric circuit when closed. When the door or window is opened, and the contact between the two magnets is broken, the circuit is also broken. This sends a signal to the security alarm's controller, which interprets this as a possible unexpected entry.





34 percent
THE PROPORTION OF **BURGLARS** WHO ENTER THROUGH THE **FRONT DOOR**

CONTROL PANEL

An alarm system's controller enables the user to arm or disable the system by entering a specific numeric code. This central control point can also allow the user to enable only the security systems within certain zones or rooms. When armed, the controller monitors data sent by the sensors and, if triggered, sounds alarms, deploys any electronic locks, and may use wireless links to alert security guards or the police.

